

H524 Introduction to Biostatistics

Week 1: Introduction to R, Data Presentation, and Summary Measures

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Outline

- 1 Course Introduction
- 2 Introduction to R
- 3 Data Presentation
- 4 Numerical Summary Measures
- 5 Rates and Standardization
- 6 AI as a Learning Tool
- 7 R Examples

- **Course:** H524 Introduction to Biostatistics
- **Credits:** 4 credit hours
- **Format:** Online
- **Schedule:**
 - Lecture: Monday, Wednesday 12:00-1:20pm
 - Lab: Wednesday 8:00am-9:50am
- **Textbook:** Principles of Biostatistics, 2nd edition (Pagano & Gauvreau)

By the end of this course, you will be able to:

- 1 Select and generate graphical and numerical summaries of data
- 2 Use principles of statistical inference to make conclusions about populations from samples
- 3 Communicate statistical findings to others
- 4 Use computer software to conduct statistical analysis

Evaluation

Component	Weight
Homework	30%
Midterm Exam	30%
Final Exam	30%
Class Participation	10%

Why R for Biostatistics?

- **Free and Open Source:** No licensing costs
- **Comprehensive:** Extensive statistical capabilities
- **Reproducible Research:** Scripts can be saved and shared
- **Graphics:** Excellent data visualization capabilities
- **Community:** Large user base with extensive documentation
- **Packages:** Specialized biostatistics packages available

R Basics: Getting Started

- **Console:** Interactive command line
- **Scripts:** Save your work in .R files
- **Working Directory:** Know where your files are located
- **Help System:** Use `?function_name` or `help(function_name)`

Basic commands:

- `getwd()` - Check working directory
- `setwd()` - Set working directory
- `ls()` - List objects in workspace
- `rm()` - Remove objects

R Data Types

- **Numeric:** `x <- 3.14`
- **Integer:** `y <- 5L`
- **Character:** `name <- "John"`
- **Logical:** `flag <- TRUE`
- **Factor:** `gender <- factor(c("M", "F", "M"))`

Data structures:

- **Vector:** `c(1, 2, 3, 4)`
- **Matrix:** `matrix(1:6, nrow=2)`
- **Data Frame:** `data.frame(age=c(25,30), name=c("A","B"))`
- **List:** `list(numbers=1:5, text="hello")`

Real datasets from published sources:

- **FEV1 (Lung Function) Data**

- Source: Pagano & Gauvreau “Principles of Biostatistics”, pp. 38-39
- Study of 13 adolescents with asthma
- Variables: FEV1 measurements (liters), gender

- **Childhood Injury Deaths**

- Source: Pagano & Gauvreau “Principles of Biostatistics”, pp. 24-25
- 100 children aged 5-9 who died from injuries, 1980-1985
- Variables: Cause of death (categorical)

Additional sources available:

- Serum cholesterol: National Health Examination Survey (pp. 12-15)
- Birth weight: 3.7 million US births, 1986 (pp. 26-28)
- All data publicly available through government agencies and published studies

- **Quantitative (Numerical Data)**

- *Continuous*: Can take any value within a range
 - Blood pressure (120.5 mmHg)
 - Body temperature (98.6°F)
 - Cholesterol level (185.3 mg/dL)
 - Time to recovery (3.5 days)
- *Discrete*: Countable values
 - Number of hospitalizations (0, 1, 2, 3...)
 - Pill count in a bottle (30, 60, 90...)
 - Number of symptoms reported (1, 2, 3...)
 - Children in family (0, 1, 2, 3...)

- **Qualitative (Categorical Data)**

- *Nominal*: No natural ordering
 - Blood type (A, B, AB, O)
 - Gender (Male, Female, Other)
 - Insurance type (Private, Medicare, Medicaid)
 - Medication name (Aspirin, Ibuprofen)
- *Ordinal*: Natural ordering exists
 - Pain scale (None, Mild, Moderate, Severe)
 - Disease stage (Stage I, II, III, IV)
 - Education level (Less than HS, HS, College, Graduate)
 - Smoking status (Never, Former, Current)

Frequency Distributions

Purpose: Organize and summarize data

For Categorical Data:

- Frequency tables
- Relative frequency (proportions)
- Percentages

For Continuous Data:

- Group data into intervals (bins)
- Create frequency tables
- Choose appropriate number of intervals

Graphical Displays: Categorical Data

● Bar Charts

- Bars separated by spaces
- Height represents frequency or percentage
- Good for comparing categories

● Pie Charts

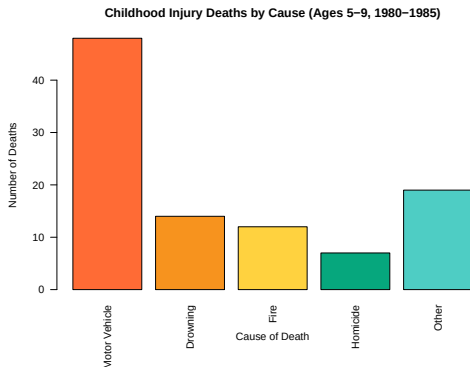
- Shows proportions of a whole
- Best for few categories
- Less effective for precise comparisons

R Example:

```
# Create bar chart for education levels
education_counts <- table(survey_data$education)
barplot(education_counts, main = "Education Distribution",
        ylab = "Frequency", col = c("#FF6B35", "#F7931E"))
```

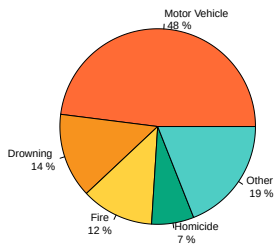
```
# Create pie chart for insurance types
insurance_counts <- table(survey_data$insurance)
pie(insurance_counts, main = "Insurance Distribution")
```

Bar Chart Example



Pie Chart Example

Childhood Injury Deaths Distribution (1980–1985)



Graphical Displays: Continuous Data

- **Histograms**

- Bars touch each other
- Shows distribution shape
- Data is grouped into bins (intervals)
- Bin width affects appearance and is somewhat subjective
- Different bin choices can reveal different patterns

- **Box Plots**

- Shows median, quartiles, outliers
- Good for comparing groups
- Compact display

- **Dot Plots**

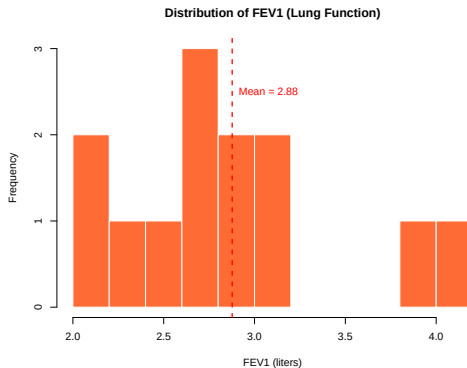
- Each observation as a dot
- Good for small datasets

R Example:

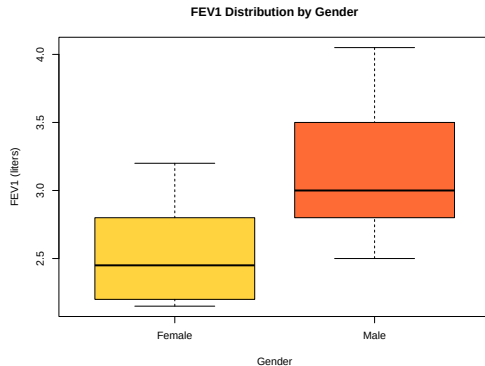
```
# Histogram for BMI distribution
hist(health_data$bmi, main = "BMI Distribution",
     xlab = "BMI", col = "#FF6B35")

# Box plot for blood pressure
boxplot(health_data$systolic_bp, main = "BP Distribution")
```

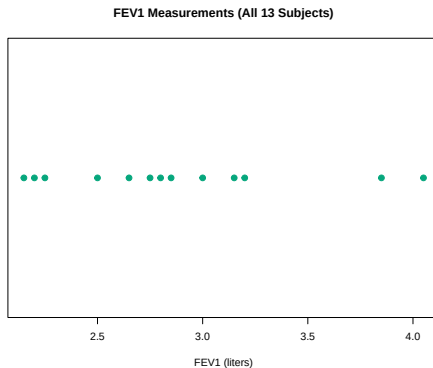

Histogram Example



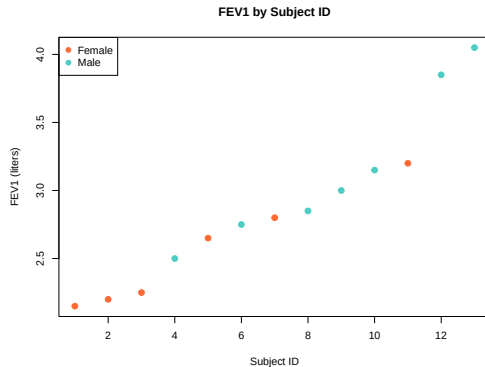
Box Plot Example



Dot Plot Example



Scatter Plot Example



Measures of Central Tendency

- **Mean ($\bar{x}$)**

- $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$
- Sensitive to outliers
- Most common measure

- **Median**

- Middle value when data is ordered
- Resistant to outliers
- Good for skewed distributions

- **Mode**

- Most frequently occurring value
- Can have multiple modes
- Useful for categorical data

Measures of Variability

- **Range**

- Maximum - Minimum
- Simple but sensitive to outliers

- **Variance (s^2)**

- $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$
- Average squared deviation from mean

- **Standard Deviation (s)**

- $s = \sqrt{s^2}$
- Same units as original data

Percentiles and Quartiles

- **Percentiles:** Value below which a certain percentage of data falls
- **Quartiles:** Divide data into four equal parts
 - Q1 (25th percentile): First quartile
 - Q2 (50th percentile): Median
 - Q3 (75th percentile): Third quartile
- **Interquartile Range (IQR):** $Q3 - Q1$
 - Contains middle 50% of data
 - Resistant to outliers

Introduction to Rates

Rate: A measure of the frequency of occurrence of an event

General form:

$$\text{Rate} = \frac{\text{Number of events}}{\text{Population at risk}} \times k$$

where k is a constant (often 1,000, 10,000, or 100,000)

Why use rates?

- Enable comparison between populations of different sizes
- Standardize measurements across time periods
- Account for population at risk

Common Types of Rates

- **Crude Birth Rate**

$$\frac{\text{Number of births in a year}}{\text{Total population}} \times 1000$$

- **Crude Death Rate**

$$\frac{\text{Number of deaths in a year}}{\text{Total population}} \times 1000$$

- **Disease Incidence Rate**

$$\frac{\text{New cases of disease}}{\text{Population at risk}} \times k$$

- **Case Fatality Rate**

$$\frac{\text{Deaths from disease}}{\text{Total cases of disease}} \times 100$$

Age-Specific Rates

Problem: Crude rates can be misleading when comparing populations with different age structures

Solution: Calculate rates for specific age groups

$$\text{Age-specific rate} = \frac{\text{Events in age group}}{\text{Population in age group}} \times k$$

Advantages:

- More precise comparisons
- Accounts for age distribution differences
- Reveals age-related patterns

Direct Method:

- 1 Apply age-specific rates from each population to a standard population
- 2 Sum across age groups
- 3 Compare standardized rates

$$\text{Directly standardized rate} = \frac{\sum_i R_i \times P_i^{std}}{\sum_i P_i^{std}} \times k$$

where:

- R_i = age-specific rate for age group i
- P_i^{std} = standard population size for age group i

Indirect Method:

- 1 Apply standard age-specific rates to study population
- 2 Calculate expected number of events
- 3 Compare observed vs. expected

$$\text{SMR} = \frac{\text{Observed events}}{\text{Expected events}} \times 100$$

SMR = Standardized Mortality Ratio

- $\text{SMR} > 100$: Higher than expected
- $\text{SMR} < 100$: Lower than expected
- $\text{SMR} = 100$: Same as expected

AI in Biostatistics: A Modern Learning Tool

- **AI tools can enhance learning** but cannot replace understanding
- **Use AI as a statistical tutor** that requires supervision
- **Critical evaluation** of AI output is essential
- **Maintain statistical rigor** while leveraging AI assistance

Today's Goal: Learn to use AI effectively for biostatistics

Popular AI Tools:

- ChatGPT, Claude (used in today's demos), Copilot
- R-specific: GitHub Copilot for R
- Statistical help: Wolfram Alpha

AI Demo 1: Generating R Code

Prompt: “Create R code to calculate descriptive statistics for blood pressure data”

Claude AI Response:

```
# First, load the real data from our Week 1 script
source("week1_lecture_code.R")

# This creates fev1_data with FEV1 measurements from real study

# Calculate descriptive statistics for FEV1 data
mean_fev1 <- mean(fev1_data$fev1_liters, na.rm = TRUE)
median_fev1 <- median(fev1_data$fev1_liters, na.rm = TRUE)
sd_fev1 <- sd(fev1_data$fev1_liters, na.rm = TRUE)
min_fev1 <- min(fev1_data$fev1_liters, na.rm = TRUE)
max_fev1 <- max(fev1_data$fev1_liters, na.rm = TRUE)

# Display results
cat("Mean FEV1:", round(mean_fev1, 2), "liters\n")
cat("Median FEV1:", round(median_fev1, 2), "liters\n")
cat("SD:", round(sd_fev1, 2), "liters\n")
cat("Range:", min_fev1, "-", max_fev1, "liters\n")

# Alternative: use summary function
summary(fev1_data$fev1_liters)
```

What to check: Variable names, missing value handling, appropriate functions

AI Demo 2: Creating Visualizations

Prompt: “Create a professional histogram of blood pressure data with normal curve overlay”

Claude AI Response:

```
# Assumes bp_data is already loaded (from week1_lecture_code.R)
# Create histogram with normal curve overlay
hist(bp_data$systolic_bp,
      breaks = 15,
      probability = TRUE, # Use density instead of frequency
      main = "Distribution of Systolic Blood Pressure",
      xlab = "Systolic BP (mmHg)",
      ylab = "Density",
      col = "lightblue",
      border = "white")

# Add normal curve overlay
x_vals <- seq(min(bp_data$systolic_bp), max(bp_data$systolic_bp), length = 100)
mean_val <- mean(bp_data$systolic_bp)
sd_val <- sd(bp_data$systolic_bp)
y_vals <- dnorm(x_vals, mean = mean_val, sd = sd_val)
lines(x_vals, y_vals, col = "red", lwd = 2)

# Add vertical line for mean
abline(v = mean_val, col = "darkred", lwd = 2, lty = 2)
legend("topright", c("Normal Curve", "Mean"),
      col = c("red", "darkred"), lwd = 2, lty = c(1, 2))
```

AI Demo 3: Explaining Statistical Concepts

Student Question: “When should I use median instead of mean for blood pressure data?”

Claude AI Response:

- **Use median when:**

- Data has outliers (very high/low BP readings)
- Distribution is skewed (not bell-shaped)
- Want robust measure unaffected by extreme values

- **Use mean when:**

- Data is approximately normally distributed
- No significant outliers present
- Need measure that uses all data points

- **For blood pressure:** Often use median because BP can have extreme values due to measurement errors or acute conditions

Always verify: Check your actual data distribution before deciding!

AI Demo 4: Interpreting Results

Prompt: “Interpret these blood pressure analysis results for a health class”

Summary Statistics:

Mean: 133.6 mmHg, SD: 18.6 mmHg

Median: 131.0 mmHg, Range: 96-196 mmHg

Males: Mean = 136.2 mmHg, Females: Mean = 130.8 mmHg

Claude AI Interpretation:

- **Central tendency:** Average BP (133.6) is in “high normal” range (AHA guidelines: ≤ 120 normal, 120-129 elevated, 130+ high)
- **Variability:** Large SD (18.6) indicates substantial individual differences
- **Distribution:** Mean $\neq$ median suggests slight right skew (some high outliers)
- **Gender difference:** Males average 5.4 mmHg higher (typical pattern)
- **Range:** Wide range (96-196) includes both hypotensive and severely hypertensive readings

Clinical relevance: About 30% of this population likely has hypertension requiring medical attention

AI Best Practices for Biostatistics

DO:

- Start with specific, clear prompts
- Ask for explanations of code
- Request multiple approaches
- Verify results independently
- Ask for biological context
- Use AI to check your work

DON'T:

- Blindly copy-paste code
- Skip understanding the logic
- Use without domain knowledge
- Ignore warning messages
- Accept first response
- Replace critical thinking

Golden Rule: AI should enhance your learning, not replace it

Common AI Pitfalls in Statistics

- **Outdated Information:** AI training data may not include latest statistical methods
- **Context Confusion:** May suggest inappropriate tests for biomedical data
- **Assumption Violations:** Doesn't always check statistical assumptions
- **Missing Values:** May not handle missing data appropriately
- **Interpretation Errors:** Can confuse correlation with causation
- **Package Dependencies:** May suggest functions from packages you don't have

Always:

- Check your data fits the suggested method
- Verify assumptions are met
- Cross-reference with textbooks/course materials
- Test code on small examples first

Effective Prompting for Biostatistics

Poor Prompt: ‘‘Help with statistics’’

Better Prompt: ‘‘I have blood pressure data from 100 patients. Create R code to compare BP between males and females using appropriate statistical test.’’

Best Prompt:

‘‘I have blood pressure data from 100 patients with variables: age, gender, systolic_bp, treatment_group. I want to:

1. Compare systolic BP between males and females
2. Check if data meets normality assumptions
3. Choose appropriate statistical test
4. Interpret results in clinical context
5. Create publication-quality plot

Please provide R code with explanations.’’

Key elements: Context, sample size, variables, specific goals, level of explanation needed

AI Assignment Preview

This week's homework will include:

- ➊ Using AI to generate R code for data analysis
- ➋ Comparing AI output with manual coding
- ➌ Identifying and correcting AI errors
- ➍ Writing effective statistical prompts
- ➎ Critical evaluation of AI explanations

Learning Goals:

- Develop AI literacy for biostatistics
- Maintain statistical rigor while using AI tools
- Build confidence in statistical computing
- Prepare for modern data science workflows

Remember: AI is a powerful tool, but you are the biostatistician!

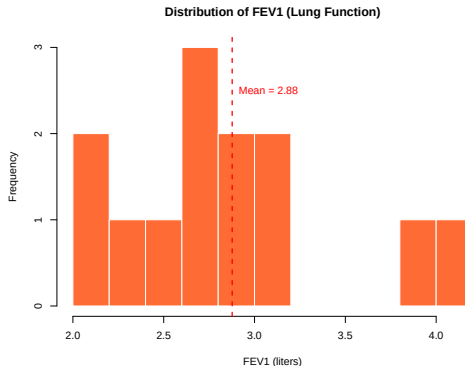
R Example: Blood Pressure Study Data

```
# To follow along: source the complete script
source("week1_lecture_code.R")

# Or generate the data yourself:
set.seed(123)
bp_data <- data.frame(
  age = round(rnorm(100, mean = 45, sd = 15)),
  gender = sample(c("Male", "Female"), 100, replace = TRUE),
  systolic_bp = round(rnorm(100, mean = 130, sd = 20)),
  treatment = sample(c("Control", "Treatment A", "Treatment B"),
    100, replace = TRUE)
)

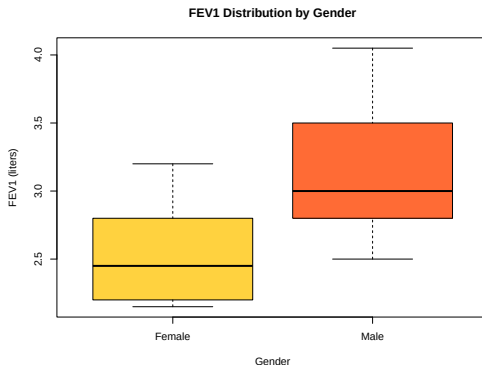
# Basic descriptive statistics
mean(bp_data$systolic_bp)      # Mean: 133.6 mmHg
sd(bp_data$systolic_bp)       # SD: 18.6 mmHg
summary(bp_data$systolic_bp)  # Five-number summary
```

Data Visualization: Histogram



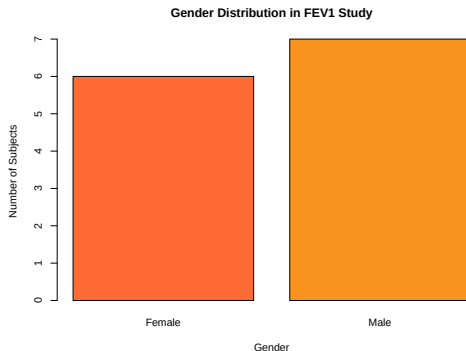
- Shows distribution of FEV1 lung function measurements
- Red dashed line indicates mean value (2.88 liters)
- Real data from 13 adolescents with asthma

Data Visualization: Box Plots



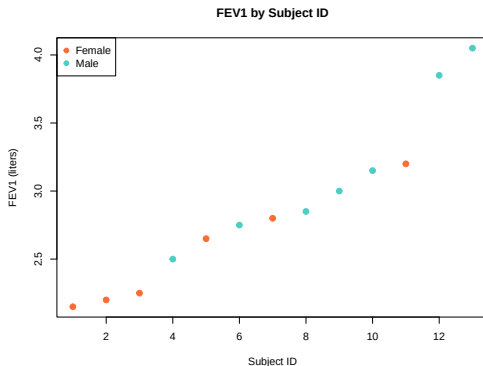
- Compares FEV1 distributions between genders
- Shows median, quartiles, and outliers
- Males have higher median FEV1 than females

Data Visualization: Bar Charts



- Good for categorical data
- Shows frequency counts for gender in FEV1 study
- 6 females, 7 males in the study

Data Visualization: Scatter Plots



- Shows FEV1 by subject ID, colored by gender
- No clear pattern with subject ID
- Illustrates gender differences in lung function

R Code: Creating Visualizations

```
# Histogram
hist(bp_data$systolic_bp,
     main = "Distribution of Systolic Blood Pressure",
     xlab = "Systolic BP (mmHg)", col = "lightblue")

# Box plot by gender
boxplot(systolic_bp ~ gender, data = bp_data,
       main = "Systolic Blood Pressure by Gender",
       col = c("pink", "lightblue"))

# Scatter plot with regression line
plot(bp_data$age, bp_data$systolic_bp,
     xlab = "Age (years)", ylab = "Systolic BP (mmHg)")
abline(lm(systolic_bp ~ age, data = bp_data), col = "red")
```

R Example: Rates Calculation

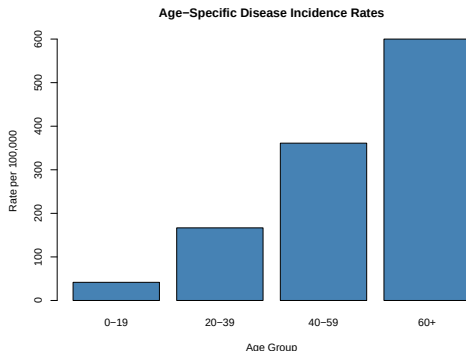
```
# Disease surveillance data
population_size <- 50000
new_cases <- 125
deaths <- 15

# Calculate incidence rate
incidence_rate <- (new_cases / population_size) * 100000
# Result: 250 per 100,000 population

# Calculate case fatality rate
case_fatality_rate <- (deaths / new_cases) * 100
# Result: 12%

# Age-specific rates
population_by_age <- c(12000, 15000, 18000, 5000)
cases_by_age <- c(5, 25, 65, 30)
age_specific_rates <- (cases_by_age / population_by_age) * 100000
```

Age-Specific Rates Visualization



- Clear age gradient in disease incidence
- Highest rates in oldest age group (600 per 100,000)
- Demonstrates importance of age-specific analysis

Today we covered:

- Introduction to R and its capabilities
- Types of data and appropriate presentation methods
- Graphical displays for different data types
- Numerical summary measures (central tendency and variability)
- Rates and age standardization methods

Next week: We'll dive deeper into probability concepts

Reading: Pagano & Gauvreau, Chapters 1-4

Thank you!

Office Hours: Wednesday, 10:00 am – 11:30 am

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